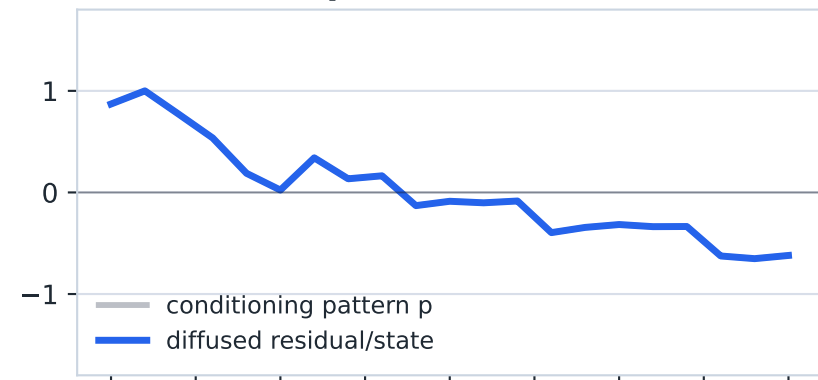


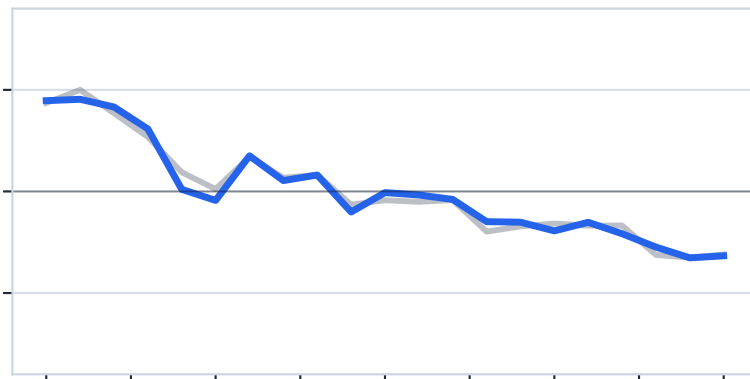
Forward Diffusion: Clean Pattern Signal Is Gradually Mixed With Noise

Visualized with a learned GOOG K=11 SISC centroid and the paper-code beta schedule: $x_t = \sqrt{\alpha_{\text{bar}_t}} x_0 + \sqrt{1-\alpha_{\text{bar}_t}}$ noise.

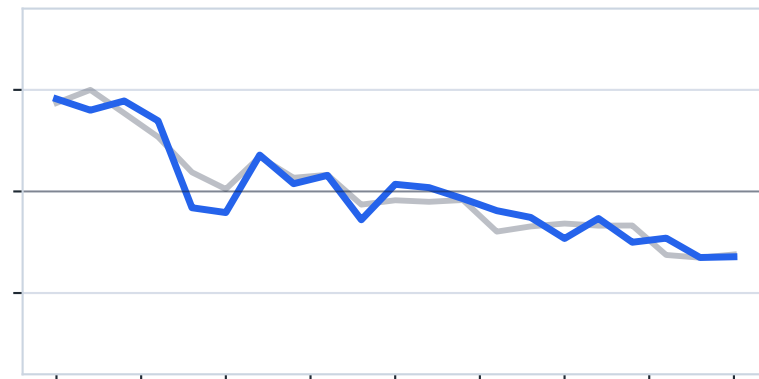
Diffusion step t=0



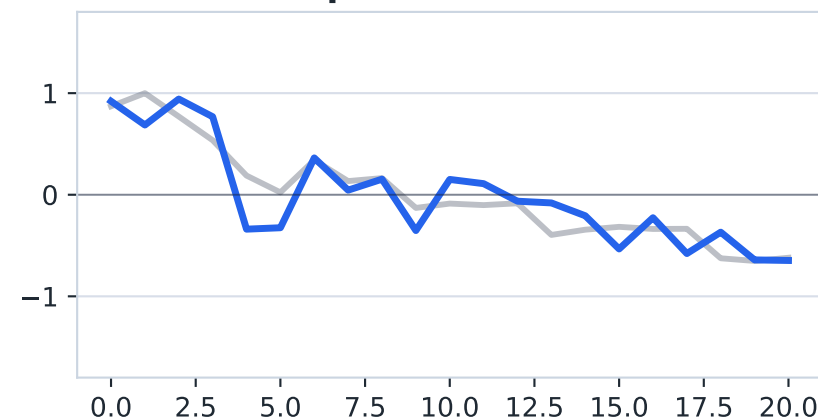
Diffusion step t=4



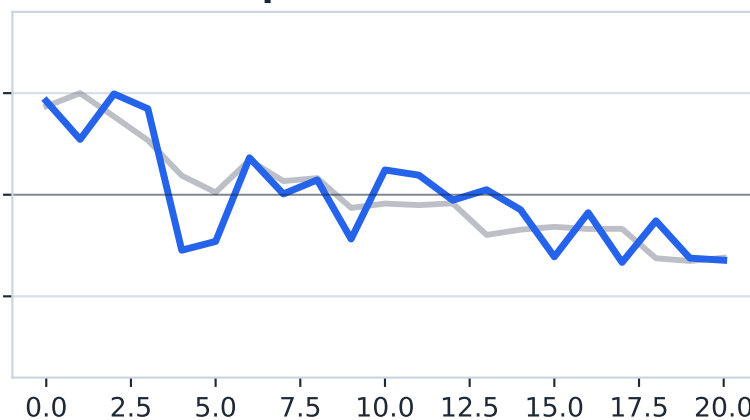
Diffusion step t=9



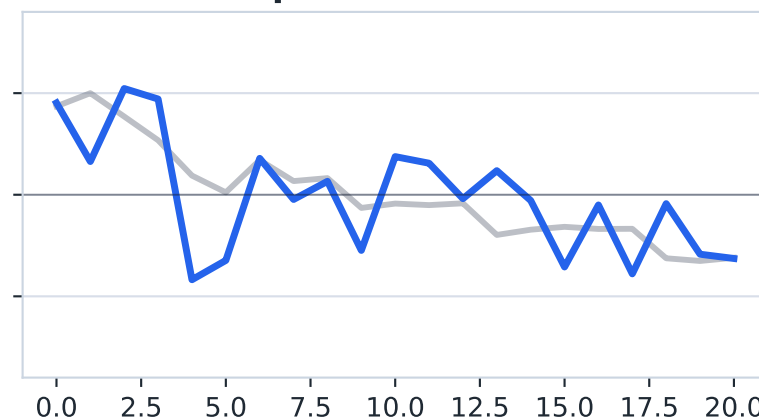
Diffusion step t=14



Diffusion step t=20



Diffusion step t=29



This is the exact forward noising equation used by the pattern-conditioned diffusion module; the sampled noise is fixed across panels.